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This form of the limiting function *F(T)* has been used over several years; however, alternative functions could be considered, provided that the value of *F(T)* approaches 100% or 1.00 as *T* increases – this implies that, over the long term average, the amplifier can not be run at over 100% of its allowable stress.

Example: A typical *F(T)* curve might be

$$F(T) < \underset{\text{I}}{\text{minimum}} \begin{cases} 12.5 \\ 50/T + 2 \\ 500/T + 1 \end{cases}$$

The figure below graphs the three terms, and the "minimum" curve. There are two breakpoints, one at *T*=4.76ms, *F*=12.5 and a second at *T*=450ms, *F*=2.11.

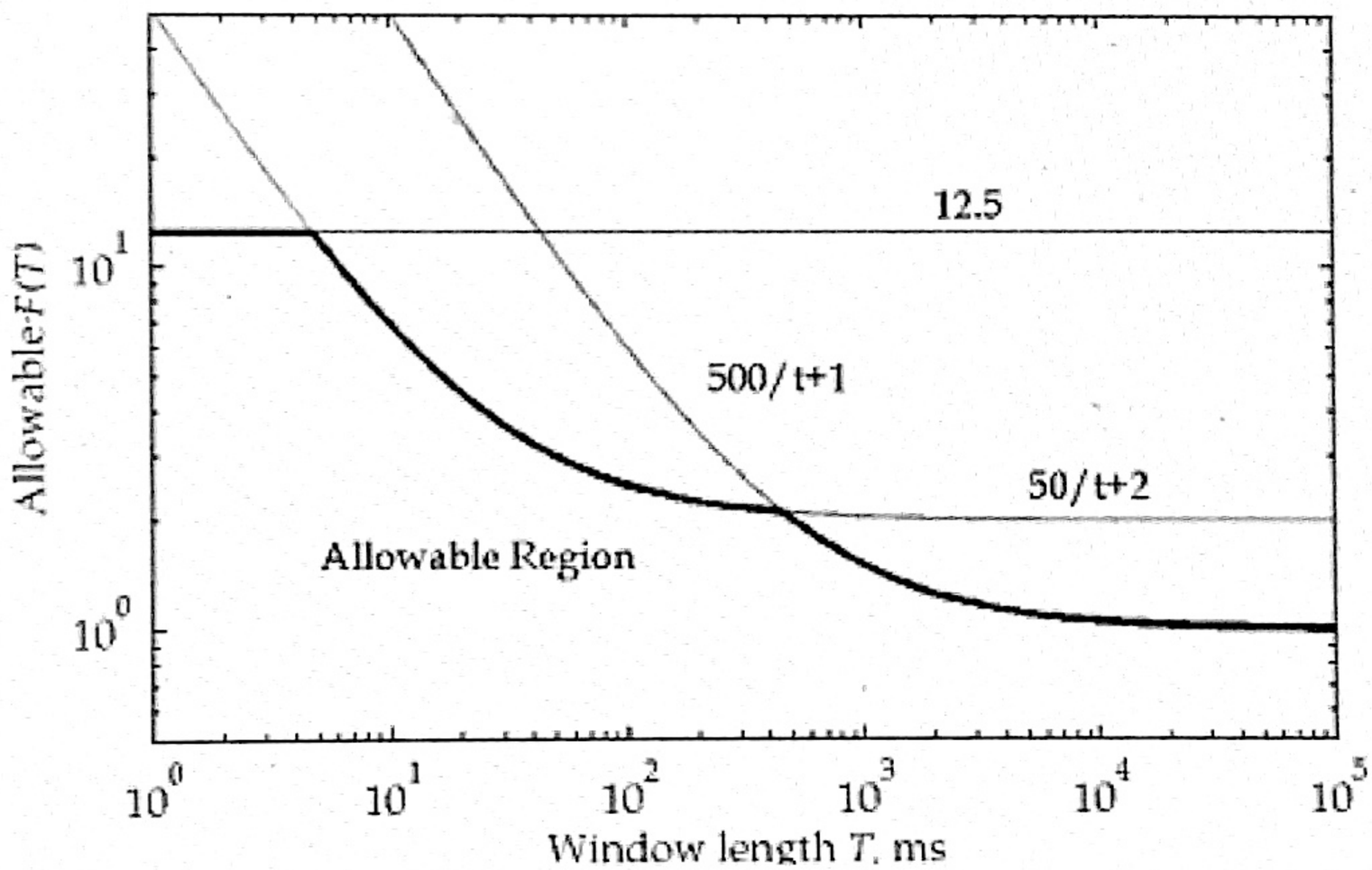


Fig. 4-9: Typical windowed averaging limit curves

If we use the stress function *f(t)* from the last section, the maximum value is 12.5 at full power. The *F(T)* curves say that the longest full power rectangular pulse available is 4.76ms; this is given by the upper corner point. If this 4.76ms pulse is repeated every 28.2ms, the average of *f(t)* is *F*=2.11 (which was chosen to match the lower corner point). Since up to 450ms is allowed at 2.11, a sequence of 19 pulses can be used. This sequence can be repeated every 1,1230ms, which will bring the long term *F* down to 1.00.

4.2.2.3 Gradient coil power dissipation

The gradient coil power dissipation system safety limits are in place to limit the GC temperature, by limiting power dissipation within the gradient coil. In turn, this limits the heat generated by the coil, and thus the temperatures.

The gradient coil SSL protects the coil itself; internal temperatures must be limited to prevent coil damage. In addition, since gradient coil inner diameter surface temperatures are a major factor in determining bore wall temperature, the GC SSL plays a critical role in managing bore wall temperature.

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The SSL parameters are kept in a configuration file, as they vary with different coil designs.

4.2.2.3.1 Calculating GC dissipation

System safety limits are placed on the dissipation within any one axis (P_x, P_y, or P_z), and on the total dissipation within the coil, P_{GC}. The total GC power dissipation is assumed to be the sum of the three individual axes' dissipations,

$$P_{GC} = P_x + P_y + P_z \tag{4.7}$$

Each axes' dissipation is calculated using GC resistances R_x, R_y, and R_z and the gradient current waveforms I_x, I_y, and I_z. Because the coil resistance R depends on frequency, i.e. $R = R(f)$, the dissipations are usually calculated in the frequency domain.

The dissipation in any individual axis is defined as

$$P = \int_{-\infty}^{\infty} I^2(f)\,R(f)\,df \tag{4.8}$$

Since typical MR gradient waveforms are repetitive, the integration converts to a summation of discrete frequencies. The frequency range of the summation can be limited to ±10 KHz, as gradient waveforms have little energy at higher frequencies. The dissipation is then

$$P = \sum_{n=-N}^{N} I^2(n\Delta f)\,R(n\Delta f) \tag{4.9}$$

where $I(n\Delta f)$ is obtained by the DFT (or FFT) of the gradient waveform.

Characterization of the gradient coil resistance $R(f)$ is covered in section 4.6.3.5.

4.2.2.3.2 Burst mode considerations

Generally, the gradient coil has thermal time constants that are longer than a typical scan; typically, they are on the order of minutes. The thermal time constants vary depending on coil design, and they also are different for different locations in the coil.

The GC's time constants can be exploited to allow scans to run at GC dissipations higher than the "steady state" or continuous limits, provided that there is idle or low dissipation scans interspersed in the overall sequence of scans to allow for GC cooling. However, care must be taken to ensure that long term limits are adhered to. Otherwise, overheating will result.

At present, the system design does not exploit this thermal behavior; higher GC dissipation than the steady state limit is not allowed.

4.2.2.4 Gradient driver capability

The gradient driver system safety models limit the pulse sequence repetition time such that the steady state temperature limits of the electronic modules on the gradient driver will not be violated. A summary will be covered in this section. For more details, refer to the Rio Gradient Driver Capability Model (DOC1586704).

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4.2.2.4.1 IGBT temperature

Due to the large amount of current flowing through gradient driver (and hence IGBTs and diodes), significant amount of heat is produced resulting in significant temperature increase. This model predicts the temperature rise on each of these components and limits the pulse sequence repetition time such that none of the components exceed the steady state maximum temperature limit (125 C).

4.2.2.4.2 RMS current

The cables and electronics components of the gradient driver are rated for a maximum current (I_{amax})at steady state. The objective of this model is to limit the pulse sequence repetition time such that the steady state RMS current limit of the cables and electronics will not be violated. The RMS current is calculated for a given time period, T, using

$$I_{rms} = \sqrt{\frac{1}{T} * \int_0^T I_{coil}^2(t)\, dt}$$

If the calculated value is greater that the RMS current limit, then deadtime is added to the sequence using the square of the ratio of the calculated current to the current limit.

4.2.2.4.3 Bus bar temperature

The objective of the Bus Bar thermal model is to limit the pulse sequence repetition time such that the steady state temperature limits (91 C) of the XGD driver bus bar will not be violated.

4.2.2.5 Gradient power supply capability

The objective of the power supply capability model is to limit the pulse sequence repetition time such that steady state power limits of the power supply and power drive unit will not be violated.

4.2.3 Prescan

The system needs to adjust, or self-calibrate, to adapt to variations between patients (size, position, anatomy) and between scans (transmit field strength, receive signal strength, FOV, local coils used, etc.). This pre-scan calibration process – usually just called *prescan* – typically determines

1) RF center frequency,
2) transmit power,
3) receive gain,
4) first-order shim settings.

For some scans, additional calibrations may be required, such as

1) phase correction (FSE),
2) I/Q transmit settings (dual drive systems in optimize mode).

This section gives a high-level overview of the purpose and implementation of these functions. For more information, refer to the Prescan Processing SWDD (DOC0955115). Note that Lx, Mx, host and spectral prescan algorithms/processes are separate entities.

The system provides a mode that allows the reuse of system prescan parameters to minimize the amount of time setting up the system for the patient and application. This mode is referred to as smart prescan. Each prescan parameter has rules determining what triggers redetermination.

The system is designed to detect if there is not sufficient MR data to complete the following prescan processes: CFL, TG, CFH, Receiver Gain, and FSE phase correction. The system examines the peak magnitude value of the incoming data frame to determine the signal intensity.

The general order (without smart prescan) is CFL, TG, autoshim, CFH, receiver gain, receiver noise, FSE phase correction (if applicable) then I/Q transmit settings (if a dual drive in optimize mode).

4.2.3.1 Center frequency (CFL)

The system data acquisition is designed such that it assumes the center frequency, CF, is set to match the Larmor frequency of the water peak. CF may vary between scans because of magnet B0 drift, FOV location, patient physiology, and other effects. The default CF value for the system may be set either in the Manual prescan window (Save CF function) or set by the LVShim tool.

The center frequency low resolution (CFL) prescan process uses a large frequency bandwidth (2kHz for 1.5T, 4kHz for 3T) to be able to locate the water resonance peak relative to the default center frequency. A larger bandwidth allows the process to locate a peak that is farther from the current saved value. The input signal is gathered from the slice closet to the center of the imaging volume.

For smart prescan, this process is re-executed if a new landmark is set or a different RF coil is used.

4.2.3.2 Transmit gain (TG)

The system sets the *transmit gain* (TG) by changing the RF pulse amplitude at the exciter output. In turn, this changes the output power level of the RF amplifier. TG is described as 0 to 200 counts, with each count equal to 0.1dB, and TG=200 being full scale transmit power. By extension, TG=0 is 20dB below full scale power or 1% of full power. The transmit attenuation, TA, is given by the equation

$$TA = 200 - TG \tag{4.10}$$

In all DV software releases, there are two methods employed to set TG, fast TG and slow TG. DV24 has included a third method, xTG, which takes less time to complete than fast TG. In the current design, the slow TG method is used if either of the fast TG methods fails to complete successfully.

For the slow TG method, a series of spin echo scans are performed without phase encoding gradients using a large range of TA values and calculating the optimal TA which maximizes the signal based on a Gaussian fit of the signal intensity. The resultant TA value corresponds to the RF power required to achieve a 90 degree flip. Usually 10 to 15 TA values are required to properly characterize the signal vs. TA curve.

For the fast TG method, multiple echoes from 3 RF pulses are used. The signal intensities from these RF pulses allow the flip angle to be determined at the TG value applied. Knowing the flip angle for the chosen TG value, the TG value for a 90 degree pulse can be calculated. This method only requires a single excitation and therefore takes less time to complete than the slow TG method.

The xTG method is based on the Bloch-Siegert shift. In this method, a spin echo is generated without phase encode gradients for a predefined TG value. A Fermi-shaped pulse is applied off resonance. The primarily

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observed effect of this pulse is a spin precession frequency shift. This shift is proportional to the square of the magnitude of B1. Knowing the B1 related to the TG value used, the TG value for a 90 degree pulse can be calculated.

For 1.5T systems, the slice closest to the prescribed imaging volume is collected in the prescribed orientation. For 3T systems, the slice closest to the prescribed imaging volume is collected in the axial plane. This difference was implemented to compensate for the empirical evidence indicating that if TG prescan mode is not performed in the axial plane (where the RF is the most homogeneous), the TG value is set to a value higher than desired due to large B1 falloff in the Z direction.

Finally, when the TG algorithms indicate that there is insufficient RF power to obtain the desired flip angle, the TG value is set to 190.

Note also that the transmit power actually used for data acquisition is not necessarily the value presented to the user. The value of TA applied (TA scan) is given by

$$TA\ scan = TA\ user + xmtaddAPS1 - xmtaddScan, \tag{4.11}$$

where *xmtaddAPS1* is associated with the prescan TG method and *xmtaddScan* is associated with the application being used (e.g. FSE).

4.2.3.3 Autoshim (AS)

The 3-plane Autoshim acquisition measures and allows correction of low spherical harmonic order in vivo frequency variations. Autoshim acquires two phase images, one with fat-water in-phase, the other with fat-water out of phase. A phase difference map is generated by subtraction. The linear and quadratic shim terms are calculated using least-square curve fitting to the derivative of the phase difference image. The linear terms are converted into DC current sent to the gradient coils.

4.2.3.4 Center frequency fine (CFH)

The center frequency high resolution (CFH) prescan process refines the value of the center frequency value to identify the water resonating peak to a higher precision than CFL. A smaller receiver bandwidth (500Hz for 1.5T and 1000Hz for 3T) is used to collect the spectral information. Input signal is gathered from the slice closet to the center of the imaging volume. An inversion recovery (IR) pulse is applied to suppress the fat signal.

4.2.3.5 Receive gain (R1, R2)

The receiver gain (RG) prescan process adjusts the amount of RF signal from the patient reaching the receiver. The range of analog RF that can be converted to digital is limited. Maximum and minimum is determined by the dynamic range of the signal. The amount of RF reaching the receiver is regulated by 2 filters: analog amplification (R1) and digital amplification (R2). The process sets the analog (R1) and digital (R2) receive gains using the prescribed acquisition parameters to get optimal dynamic range without overranging. R1 and R2 value are set such that the expected peak signal is between 25% and 50% of the maximum dynamic range.

To set the R1 value, prescan plays out the center view of k-space; since this view has no phase encoding, it is assumed to have the largest signal. R1 is set so that the signal from this view it is adjusted to 50% of the ADC

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full scale. A 50% target is used since the assumption of "center view gives maximum signal" may not always be true.

If the scan prescription indicates that contrast will be used with the scan, the prescan process reduces R1 by 1 count to prevent a possible ADC over-range that could occur with the higher contrast enhanced signal during the scan.

The R1 and R2 value can be manually adjusted by the slider style user control thru manual prescan.

4.2.3.6 Receiver noise

The receiver noise prescan process characterizes noise from all active receiver channels. The purpose of this process is to characterize the relative gain of the receive channels. The computed standard deviation values are referred to as "prescan channel weights" or "prescan noise weights", with each weight representing the relative gain of that channel. This process uses the maximum R1 setting as well as the extended dynamic range function. The gain factors are delivered from the reconstruction subsystem in the logical order.

4.2.3.7 Phase correction

The FSE phase correction prescan process is designed to account for uncompensated linear and B0 eddy currents. These eddy currents are major sources of phase errors in FSE and can lead to ghosting and/or signal loss. The process computes magnitude and phase difference from two frames of data and uses this information to compute the slope and intercept of the measured phase shifts. These are used by the reconstruction subsystem to correct FSE data.

Smart prescan does not impact the phase correction prescan process. Data is always collected and used for each FSE scan when the option is selected.

4.2.3.8 I/Q transmit settings

The I/Q transmit setting prescan process determines the optimum value of phase and amplitude offset of the RF signal sent to transmit channels 1 and 2 to obtain the maximum image uniformity. The range of allowed offsets for MR750w body mode is 0 to -45 degrees and 0 to +4.5 dB in attenuation amplitude. For more information about the dual drive implementation, refer to DOC0932542 (Dual Drive Feature Design Document).

4.2.4 High Order Shim

High order shimming, or *HOS*, is *B0* shimming done on a patient-by-patient basis. It improves *B0* homogeneity by compensating for the patient-induced effects on the *B0* field.

The "high-order" name is somewhat misleading, for two reasons. First, HOS adjusts both the low order shims (by adjusting gradient current offsets) and higher order shims (generally by driving currents into a set of shim coils). Second, even basic magnet shimming, done during installation and field service, has controls for high order shims.

The term "HOS" as used in this section covers the shim adjustment system used on a per-patient basis applying current through a set of dedicated resistive coil windings.

There are several components to HOS:

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1. a method to measure shim, over a user-controllable region of interest
2. a mechanism to adjust shim, usually a combination of gradient current offsets and controlled shim currents into a set of shim windings
3. a user interface to integrate the first two into the operator's workflow.

In this document, the content will be limited to driving body-sized spherical harmonic resistive windings at a constant current. Although being examined in the research community, the support of slice-specific shim settings, pre-emphasis to reduce settling time, or driving a number of loop coils (matrix shimming) has not been implemented in GE products and is not in scope.

4.2.4.1 HOS driver design

The requirements for a high order shim driver flow from the system requirements of number of spherical harmonic (SH) terms/windings, the maximum field for each SH winding and the maximum total field. The number of SH terms dictate the minimum number of independent channels are required. The maximum field requirement in conjunction with the HOS coil gain requirements, dictate the required current per channel. The maximum field requirement dictates the total current required from the driver. The design of the driver is impacted strongly by the HOS coil design. The resistance of the HOS coils will dictate how much voltage is required to deliver the maximum current. The coupling of the windings internal to the gradient coil dictate how much voltage headroom is required to overcome induced voltage on the HOS windings and still deliver the maximum current. The coupling of the gradient windings also dictates the design of the driver control circuit. An element of the driver design not related to the HOS coil design is communication: the protocol used to accept demands to change current as well as to report status.

To date HOS drivers have been linear amplifiers (as opposed to switching amplifiers). These drivers have a limited range of supported resistive loads bounded at the low end by the thermal properties of the amplifier and bounded at the high end by the maximum available voltage.

4.2.4.2 HOS coil design

As with the HOS driver, the requirements for the HOS coil(s) flow from the system requirements of number of spherical harmonic (SH) terms/windings, the maximum field for each SH winding and the maximum total field. The most important design elements of the HOS coils are the gain (field per amp), the resistance and the orthogonality of the windings. The gain of the coils will dictate the maximum field produced at a given current. The resistance of the coils dictates the heat that will be generated passing current through them. The HOS windings must be able to support the maximum current provided by the driver. This includes addressing the heat generated by the IR drop.

Until recently, HOS coil design requirements were written in terms of purity. That is, the design targeted creating pure spherical harmonic fields. If the coils were pure, this also implied that they were orthogonal. Supplying the five second-order SH coils implied that all second-order fields could be compensated. Recently, after an update to the EM design software, the design requirements have replaced purity with orthogonality. Although the winding target field patterns main components follow the spherical harmonics, the resultant coils show much less interaction between themselves and between them and the imaging gradients.

Finally, the HOS coil design must ensure that the mechanical structure of the gradient coil is sufficient to remain intact during use. There have been incidents of coil structural failures in prototype gradient coils due to not allowing enough epoxy between HOS layers and/or within each layer.

4.2.4.3 HOS filter design

The HOS filter is designed to prevent RF and common-mode signals from entering the scan room. There is a filtering circuit for each HOS channel. Like the imaging gradient filter, the HOS filter is mounted to the penetration panel. The HOS filter design must consider it thermal response and provide sufficient cooling to limit the temperature of accessible surfaces.

4.2.4.4 HOS calibration tool

As its name implies, the HOS calibration tool is designed to be used by the Field Engineer to characterize the field created by each HOS winding. The tool uses a large phantom to cover a significant portion of the imaging space. An LVShim acquisition is used to characterize the field created by each coil when driving a known current through each winding individually. The tool has a simple check to verify reasonable results. The required output of the tool is field information of each coil (including the three imaging coils) that can be used by the feature to convert a desired field change into a change in current supplied to each resistive coil. At this time, the format of the tool output is a set of three field maps corresponding to three specific FOV/matrix size combinations. These are referred to as reference maps.

4.2.4.5 HOS clinical feature

The entire purpose of providing the HOS components is to provide the user with the capability to compensate for more field inhomogeneities caused by the patient in order to improve image quality. The design elements are the same as those for linear (gradient) shimming. When the option is selected, a field map or set of maps are generated, the resultant maps are masked then fit to spherical harmonic terms, the SH terms are converted to required current needed to counteract these fields, and the required DC current is applied. At this time, however, the workflow for the HOS feature is different from gradient shimming. Gradient shimming is automated and is included in prescan. The HOS feature is currently not integrated into prescan. At this time, the HOS feature workflow is as follows:

1. One of the three HOS series is included in the protocol (these HOS series correspond to the three reference maps created by the HOS calibration tool).
2. When executed, the HOS series generate 2D multislice field maps using a spiral acquisition.
3. A dedicated UI is displayed to allow the user to specify the volume of interest, calculate the target shim currents and apply the shim currents.

These shim currents will remain until the exam ends, the table moves more than a specified distance or the linear shim function is used.

4.3 Major performance parameters

4.3.1 Field strength

B0 field strength is *the* defining parameter for an MRI system. A system is often described by its field strength: "This is our 3T system." In general, signal strength is proportional to field strength, due to a higher net magnetization of tissue. Field strength has a big effect on system cost, primarily because of the magnet itself, but also because of siting cost (extent of the 5 Gauss line), and RF transmit subsystem cost.

While field strength is set by the magnet, it also has major effects on the RF subsystem complexity, especially for large anatomies (whole body imaging) at higher field strengths (3T and above). The RF amplifier power scales *roughly* with field strength. In addition, since the wavelength is similar to the extent of the human body, B1 distortion is larger at 3T than at 1.5T. Dual drive transmit has been employed to help address this.

GEHC has produced 0.2T, 0.35T, 0.5T 0.7T, and 1.0T systems in the past. Presently, the only marketed products are 1.5T and 3T. A 7T system is offered for research use; this is a whole body sized system, but with only local RF transmit capabilities – there is no RF body coil.

4.3.2 Patient bore

The patient bore size is the second most defining parameter for an MRI system. It is also a critical design decision, with increasing patient access and comfort pushing for shorter, larger diameter bore dimensions, and increased gradient and RF performance pushing for smaller bore diameters.

The "bore" diameter is the inside dimensions of the RF coil tube or "former"; these are fiberglass tubes, wound on a steel mandrel, which accurately controls the diameter. The table below summarizes the bore diameter for the DV products.

Patient bore diameter

Product	Specified	Claimed
MR450, MR740	≥600mm	60cm
MR450w, MR750w	≥695mm	70cm
Rio	≥695mm	70cm

4.3.3 Field of view (FOV)

Field of view defines the uses of the system (whole body vs. head-only or extremity-only) and is the primary factor in system cost. The three main subsystems that affect the system's FOV are:

1. the magnet, which sets the B0 homogeneous region,
2. the gradient coil, which has a limited linear region, and
3. the largest RF transmit coil (usually a whole body coil), which provides a uniform B1 field in a limited region.

Each of these is covered in the applicable subsystem section. Increasing the FOV size is expensive for each of these subsystems.

Reducing a system's FOV to a single number is an oversimplification. Some uses or applications require highly accurate fields, while others are more tolerant of inaccuracies. For example, a simple spin echo

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sequence is much more tolerant of B0 inhomogeneity than a gradient recalled echo planar imaging sequence.

At this time, there is not a clear definition for FOV. Theoretically it could be defined generally as the region with a maximum B0 field deviation, a maximum gradient field deviation and maximum B1 field deviation. Although each of these are specified for each product, they are specified independently and the specification level is not always constant. There is a practical limitation in our software that a single value is used to limit the FOV along all three axes – if three values were used, prescribing oblique slices would be much more complicated. In light of this, the FOV limit is set based on the field capability in the XY plane. For 60cm systems, this value has been set to 48cm. For 70cm systems, this value has been set to 50cm. Note that due to tradeoffs with gradient coil gain, resistance and inductance, the target imaging capability in the Z direction is less than 50cm for the 70cm systems.

4.3.4 SNR

SNR (signal to noise ratio) encompasses a wide range of performance measures. In general, a measurement result has a repeatable part (the signal) and a non-repeatable part (the noise). For repeated measurements, SNR can be defined as the ratio between the mean (μ) and the standard deviation (σ) of the measurement.

4.3.4.1 Image SNR

Image SNR is one of the most fundamental MRI system performance parameters. There are several methods of measuring image SNR; these methods, and the advantages and disadvantage of each, is involved enough to warrant an extensive discussion, which is in appendix B1, *SNR*.

Many images are constructed from multiple receive signals, with complex processes. This has caused confusion about how image SNR should be defined. However, the basic principles still apply: SNR is the ratio between the repeatable and non-repeatable measurements; it is the ratio between the mean and standard deviation for the repeated measurements.

4.3.5 Imaging speed

Imaging speed is the system's ability to gather MR data and process it, given a fixed amount of time. This encompasses a wide range of topics: minimum echo times, minimum repetition times, receive channel count, reconstruction speed, etc.

From a subsystem view, it includes gradient strength, slew rate, and average power; data acquisition bandwidths and capabilities; switching time between RF transmit and receive, etc.

4.3.5.1 Minimum TE/TR

As described previously, TE is the time between RF excitation and data acquisition and TR is the time between neighboring RF excitation of the identical slice/slab. Minimizing TE allows for the following: 1) production of image signal from short-T2 species such as tendon and bone (able to acquire signal before it has dephased) and 2) increased volume coverage in a given amount of time. Minimizing TR allows for shorter imaging times.

A 'brute force' method to reduce both TE and TR is to increase the allowable gradient slew rate and amplitude. MR signal encoding is dictated by magnetic field gradient area. A larger slew rate and amplitude capability implies less time is required to achieve the same gradient area. For RF excitation, the duration of

the RF pulse also determines the minimum TE. Generally, a shorter the RF pulse gives a degraded slice profile. For data receive, the faster the acquisition speed (higher the bandwidth), the less time required to collect a given number of slices.

There are methods to reduce TR by eliminating the requirement to wait for spins to completely relax. For example, a spoiler gradient pulse dephases the MR signal remaining after data acquisition so that the next excitation pulse can be applied. These techniques usually result in reduced SNR.

4.3.5.2 Reconstruction Speed

Imaging is considered complete when the images have been reconstructed, scaled and filtered and have been put into the image database to be viewed. The DV systems use dedicated computer system (ICNs) to perform the post-processing tasks. For most acquisition techniques, the post-acquisition processing takes a small fraction of the time it takes for data acquisition. For some techniques such as VIBRANT FLEX, post-processing is intensive. If the amount of dedicated memory is not greater than required for the incoming data, the system uses the host computer to save partially processed data, swapping between the host and ICN. This is a slow process. For these sequences, the amount of dedicated memory required goes up with the number of active receiver channels.

4.3.5.3 Partial data acquisition

Along with minimizing TE and/or TR to increase imaging speed, another technique to reduce the time required to produce the image is to reduce the amount of data acquired. Generally, the tradeoff of reduced scan time is reduced SNR. Examples of this are listed below.

1) Partial k-space: Collect a subset of phase-encode lines and use the mathematical properties of the data to complete k-space and transform to an image.

2) Partial Echo: Collect only part of the MR echo to reduce the data acquisition period and use the mathematical properties of the data to complete k-space and transform to an image.

3) Parallel Imaging: Using RF coils that have multiple independent coil elements in the encoding direction, a subset of k-space is collected and the information from the individual elements is processed to create the entire image.

4) Keyhole: To reduce the time between images when monitoring the same slices over time, only a fraction of the phase encode lines in the center of k-space are collected for all but the first image. The first image collects all of the phase encode lines. The high frequency data from the first image is then used in the subsequent data to create the images.

4.3.6 Stability

Stability is one of the most important aspects of system performance. It is also a very broad term, encompassing a wide range of performance measures.

*Generally, **stability** refers to a system's ability to get the same results if the same process is repeated.*

"Results" is a general term, covering a range of parameters: examples include the amplitude of an RF pulse, the position of an MR echo, or the image intensity of a reconstructed image. Similarly, the "process" refers to the steps and conditions used to measure the results: matching examples include repeatedly pulsing an RF

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amplifier, analyzing raw data obtained without phase encoding, or analyzing the images obtained from an fMRI sequence.

Instabilities in the system lead to loss of system performance. A basic example is loss of image quality due to ghosting or blurring of the images.

Major stability aspects are covered below.

4.3.6.1 B0 field stability

This section discusses small B0 instabilities, which are distinct from large scale instabilities, such as a magnet quench; the latter are not discussed in this document. Also, a distinction must be made between B0 instabilities and B0 eddy current effects; the latter are responses to gradient currents, and are discussed separately.

In MR systems, stability of the B0 field over time is an important performance parameter. Ideally, the magnet provides a perfectly stable B0 field. In reality, this field will change, due to many factors: environmental vibrations, cryogenic cold-head operation, heating from the gradient coil, room temperature and pressure, etc. These changes are separated into several broad categories:

1. Non-recoverable B0 drift: a loss of magnetic field strength due to energy loss in the superconducting magnet. Typically, this type of B0 drift is observed over days or weeks.
2. Recoverable B0 drift: a change in magnetic field strength due to an external stimulus to the magnet. Once the stimulus is removed, the B0 field returns to its initial condition. Typically, this type of B0 drift is observed over minutes or hours.
3. B0 fluctuations: variations in B0 field strength due to vibrations or fields, either external or internal. Typically, these are driven responses, and will decay once the external stimulation is removed. These are observed over seconds in the time domain, or up to several hundred Hz in the frequency domain.

4.3.6.1.1 Non-recoverable drift

Non-recoverable (NR) drift is a change in B0 strength over time. NR drifts accumulate over time, and require service action to reset B0 to the original value. More specifically, NR drifts are generally seen as a gradual decrease in B0 strength, observed over hours, days or weeks.

The primary and only significant source of non-recoverable drift is the magnet itself; in fact, this type of drift is often referred to as "magnet drift". B0 changes due to NR drift accumulate until the magnet is re-ramped, or "reparked". B0 strength may decay exponentially (a transient immediately after ramping) or linearly (after the initial transient has passed.)

NR drifts are usually specified as drift per unit time, either in PPM/hour or Hz/hour. NR drifts are easily measured by a magnetometer or standard system CF measurements.

During magnet production, a drift test specification of 0.1PPM/hour is considered acceptable. However, this drift rate spec is set to allow for meaningful drift testing within hours after ramping, and is much higher than the long term drift performance of a typical magnet. After any post-ramp transients have passed, typical magnets have much, much lower drift rates, usually less than 0.1PPM per *day*.

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Fig. 4–10 plots the NR drift from MR450w magnet S/N HM003 over a six month period after it was first ramped.

The measured data is modeled by a combination of an exponential decay (about 22PPM with a 28 day time constant) and a longer term linear drift (dropping at 0.04PPM/day).

Up till 80-100 days, the exponential decay dominates; after that, the linear ramp dominates. At t=0, the drift rate would be roughly 0.03-0.04 PPM/hour.

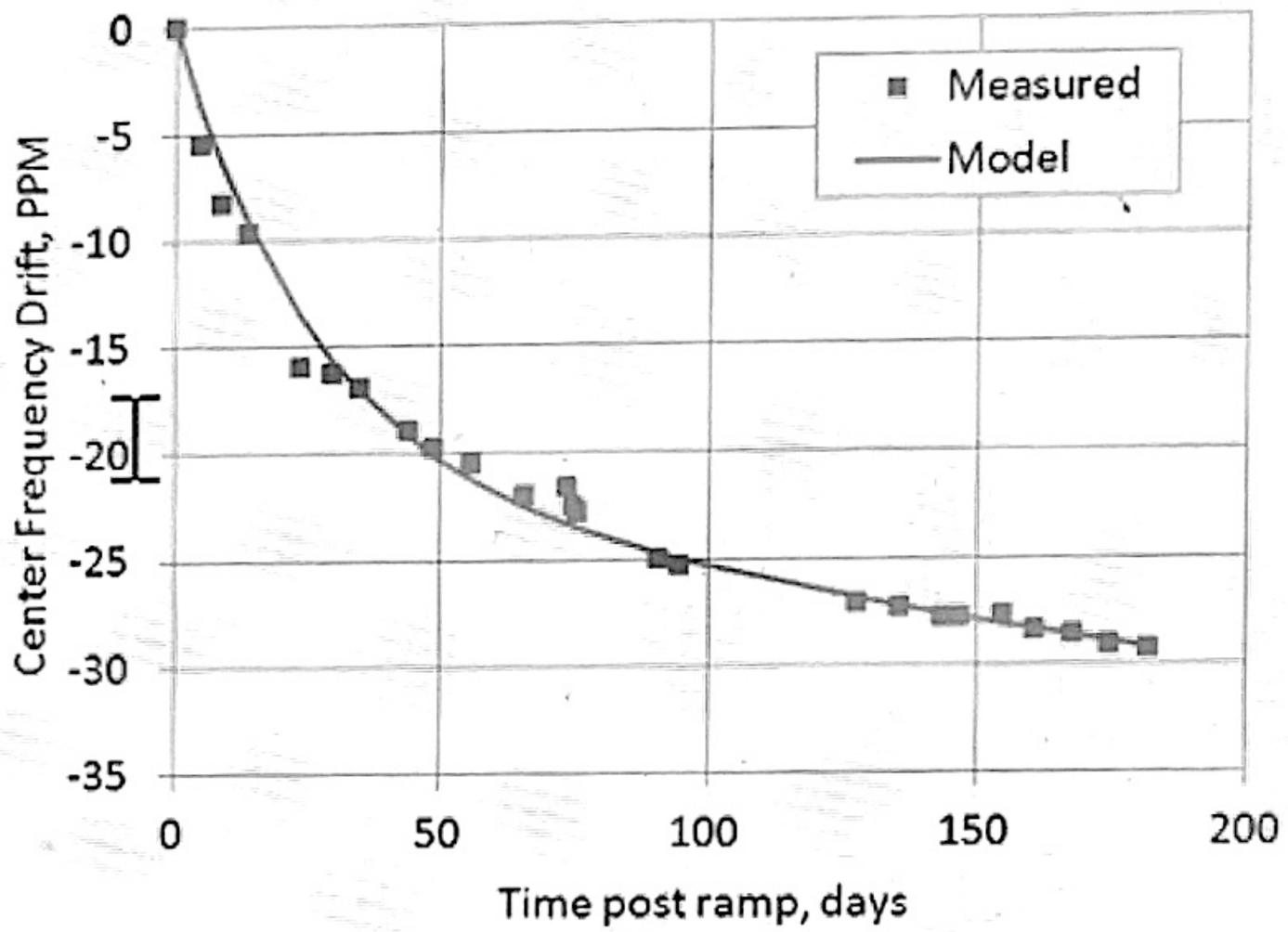


Fig. 4–10: Magnet drift, MR450w HM003

NR drift must be specified for several reasons.

1. Both spectroscopy and imaging performance are affected by NR drift. Spectroscopy is generally more sensitive; a drift of 0.02PPM during a spectroscopy experiment is detectable. Imaging is more tolerant of drift.

2. Unless NR drift is either tracked or predicted, the system may not be able to automatically and reliably find the MR signal, resulting in prescan failures. Once the magnet's Larmor frequency drifts outside of the prescan center frequency search window, prescan will fail.

MR750w (DV23.0) introduced a simple method to track the NR drift; each successful prescan logs the center frequency (CF) value, and any subsequent prescan starts the CF search at the latest successful CF value. This technique has been effective at tracking NR drift, and preventing drift related prescan failures.

4.3.6.1.2 Recoverable drift

Recoverable drift is seen as changes in B0 strength that return to the initial condition when the disturbance or cause of the drift is removed. Generally, recoverable drifts are observed over minutes or hours. There are multiple causes, including environmental temperature or pressure, heating of passive shims or magnet materials by the gradient coil, and other causes.

Recoverable drifts are generally specified by 1) the peak B0 change, in PPM, and 2) the rate of change in PPM per unit time. An alternative specification could be in units of Hz.

MR750w introduced a method to compensate for gradient heat induced recoverable B0 drift. This drift is caused by changes in the magnet and passive shim material properties, as a result of pulsing the gradient coil. The compensation method is described in section 4.2.1.4.2, *B0 drift*.

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For external disturbances (such as elevators, moving cars, moving trains, etc.) a shielding factor is specified for each magnet design. When this shielding is not sufficient, there are active compensation coils that can be installed into the site walls or one the magnet inner bore; however, these are site specific methods and not widely used.

4.3.6.1.3 B0 fluctuations

B0 fluctuations are variations in B0 field strength due to vibrations or fields, either external or internal. There are many different sources that stimulate B0 fluctuations. External sources include ventilation fans, rotating machinery, nearby traffic, etc. The main internal source is the cryogentic coldhead.

Because B0 fluctuations are driven responses, removing the stimulus eliminates the associated response. B0 fluctuations are usually characterized with the HSS tool.

4.3.6.2 Echo stability

The stability of repetitive echoes has been a major system test since the mid-1990s. Starting with TLT (Top Level Test), and evolving into SPT (System Performance Test), these *system stability* tests have proved to be invaluable in understanding and troubleshooting a wide range of system performance problems.

The principle is to obtain multiple acquisitions of "identical" echoes, and analyze the data in several ways. The choice of sequence (example: spin echo, fast spin echo, or gradient recalled echo), slice location (off-isocenter vs. at isocenter), slice orientation (which uses different combinations of gradient axes for slice, frequency and phase encoding) and scanner mode (head vs. body) can help to understand the cause of instabilities.

4.3.6.3 Image temporal stability

Image temporal stability refers to the repeatability of fMRI images, usually acquired using the EPI sequence. The tool used to characterize this performance is the fMRIQA tool which collects fMRI images of a phantom and creates a Weiskopf plot. In this method the EPI time series fluctuations described as relative deviations (Fn) are measured on the phantom and plot as a function of square ROI length. As ROI length increases the deviation of this plot from theoretical value (WP0: Fn,t=1/(n*SNR), where SNR is an image signal-to-noise ratio) characterized MRI scanner related instabilities. An example of a Weiskopf plot is shown in the figure below.

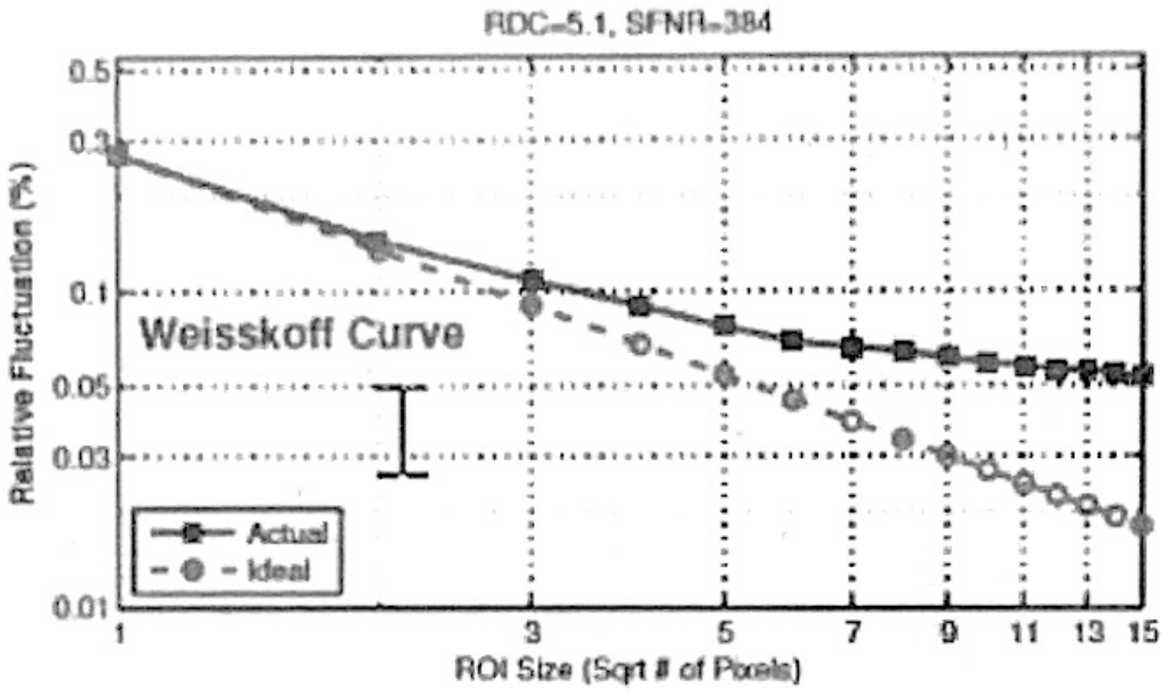


Fig. 4–11: Weiskopf plot

4.3.6.4 Gradient stability

Gradient stability is critical to MR image quality. Being able to accurately play out a waveform repeatedly allows for accurate spatial encoding of the MR spins. The gradient controller, gradient amplifiers and gradient coil contribute to the stability of the gradient waveforms. The Gradient Fidelity Diagnostic is designed to verify the accuracy, stability, and symmetry of the gradient amplifier outputs. When this diagnostic is run, a ramp-sampled EPI readout waveform is generated on each axis, separately, for 100 shots with 64 views in each shot. Gradient (current) error data is acquired, and the following tests are performed:

- *Accuracy* - Across all 100 shots, the largest point of inaccuracy is recorded, in units of microampere-seconds.
- *Shot to Shot Stability* - Each of the 100 shots is compared to the others. The largest difference between any two shots is recorded, in units of microampere-seconds.
- *View to View Stability* - Each pair of positive/negative views in the 100 x 64 views is compared to the others. The largest difference between any two views is recorded, in units of microampere-seconds.
- *EPI Plus/minus Symmetry* - Each view represented by a positive half-cycle is compared to the subsequent view represented by a negative half-cycle.

4.3.6.5 RF amplifier stability

RF amplifier stability can be separated into short-term and long-term effects, Short-term effects occur during a single RF pulse. These are characterized with the RFA tool. Short-term instabilities change the B1 field created by the transmit coil which ultimately change the flip angle distribution of the targeted spins as well as the slice/slab profile. The effect on image quality varies with PSD type but usually results in lower SNR or ghosting in the phase-encode direction.

Long-term effects are usually caused by thermal instability. The effect is usually compensated by prescan setting the transmit gain to the value to achieve a target flip angle of the spins. If gain decreases significantly, there is a possibility of not having sufficient power to achieve the required B1.

4.3.7 B1 distortion

The ideal B1 transmit field is uniform (same amplitude everywhere), rotating (to follow the spin rotation) and transverse directed (always in the *xy*-plane, assuming *B0* is aligned with *z*). If one could follow the tip of the B1 vector, it would trace out a perfect circle with each RF cycle. In the absence of a patient or other system components in the bore, a close-to-ideal field can be created using "birdcage" or related transmit coils.

Several factors, listed below, contribute to the non-ideality of the B1 field in practice.

4.3.7.1 Transmit coil effects

Birdcage coil designs, used for volume transmit coils, have some B1 uniformity limitations. These designs depend on a resonant network structure to distribute current evenly throughout the coil; uneven current distribution results in a non-uniform field. There are several effects that can disturb the resonant structure, including the asymmetry of the coil's structure, loading of the coil, and other (undesired) resonances that are close in frequency to the desired resonance. The physical dimensions of the RF coil also dictates the extent of the homogeneous RF field.

4.3.7.2 Dielectric effects

Any dielectric material in the bore will modify the transmit field. For spherical or cylindrical objects, the effect creates a higher intensity region in the center, and a gradual decrease toward the outside of the object. For non-symmetrical or elliptical objects, the effect is more complicated, and can lead to objectionable shading patterns.

Since the human body is dielectric, its presence changes the transmit field; since it is not symmetric, it leads to non-symmetric patterns. This shading is especially objectionable when the size of the body becomes similar or large compared to the RF wavelength in air. For 3T systems, the body disturbs the transmit field uniformity enough that corrections are nearly mandatory; even head imaging can be significantly improved with transmit field corrections. At 1.5T (which has twice the wavelength compared to 3T), the effects of the body are measurable, although generally do not require corrections. It is very difficult to see the effect in 1.5T heads.

The shading in elliptical or ovalized objects or patients can be reduced by modifying the transmit field. This is done by creating an elliptical transmit field that compensates for the B1 field modifications caused by the object. However, if the patient shape is more complex (like the area of the pelvis or breasts), the first-order correction available with an elliptical field only partially corrects for the effects of patient shape. Higher order correction, obtained from independent RF waveforms and/or more than two independent input channels, is required.

4.3.7.3 In-bore components

Other system components used in the bore, most noticeably local RF receive coils and their cabling, can also reduce the uniformity of the transmit field. Unlike the body, there is no intrinsic reason for this B1 distortion. The RF field (either electric or magnetic) induces current in the resonant structure (e.g. coil element, cable, etc.) which generates a field and thus alters the resultant field. Design tactics such as decoupling coil elements or cable routing are used to minimize this effect.

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4.3.8 Acoustics

The main source of acoustic noise in MR imaging is rapid alterations of currents within the gradient coils. These currents, in the presence of the strong static magnetic field of the MR system, produce significant (Lorentz) forces that act upon the gradient coils. Acoustic noise, manifested as loud tapping, knocking, buzzing, chirping, or <*your adjective here*> sounds, is produced when the forces cause motion or vibration of the gradient coils as they impact against their mountings which, in turn, flex and vibrate.

According to the IEC60601-2-33 standard, the acoustic exposure limit for the patient and MR worker is 99dB A-weighted average sound pressure. The systems are specified at a level of 126dB assuming that ear plugs will reduce the level by 27dB.

The DV product designs include several features to reduce the acoustic level in the patient bore. These include damping material on the RF coil rungs, inside the endbells, and on the ID of the gradient coil or OD of the RF body coil.

Patient comfort fans, cooling air forced between the RF body coil and gradient coil and the magnet coldhead are the main sources of ambient acoustic noise found in the MR environment. Acoustic noise produced by these subsidiary systems is considerably less than that caused by gradient magnetic fields.

The characterization method of the acoustic levels produced by the system is described in the NEMA Standards Publication MS 4. In this document, two methods are described; MGAN and MCAN. The MGAN method is designed to characterize hardware capability in worst-case conditions. For this method, the gradient coil resonances are characterized and a trapezoid gradient waveform is played out at maximum gradient current on resonance and the acoustic measurement made. When run on DV and Rio systems, the results are over the specified limit. The worst-case waveform is in general not clinically relevant. We therefore choose to use the MCAN method which is designed to characterize the maximum clinical acoustic noise. For this method, a search of the parameter space using the historically loud sequences is performed. The waveforms used for the MCAN measurement are restricted by imposed hardware limitations (mechanical lockout frequencies) as well as patient safety limits (dBdt, SAR, thermal). A report is produced for each sequence capturing the parameters used and the resultant sound pressure.

4.4 Artifacts

An image artifact is signal that does not represent the object being imaged. In general, artifacts are generated from imperfections or instabilities in the MR system. They may also be caused by intrinsic properties of the object being imaged or incorrect assumptions in the reconstruction process. For many artifacts, the impact to the image quality can be reduced or eliminated by careful selection of the scanning protocols.

A summary of some of the most common artifacts are listed below. Much of the information given below originates from a review written by L.J. Erasmus, et. al., in SA Journal of Radiology, August 2004, pp 13-17, and from mriquestions.com, created by Allen Elster, MD FACR.

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4.4.1 Motion Artifact

Periodic motion such as cardiac motion or CSF pulsation causes ghosting while non-periodic movement causes diffuse image noise in the phase-encode direction. Several methods can be used to reduce motion artifacts, including patient immobilization, cardiac and respiratory gating, signal suppression of the tissue causing the artifact, choosing the shorter dimension of the matrix as the phase-encoding direction, view-ordering or phase-reordering methods and swapping phase and frequency-encoding directions to move the artifact out of the field of interest.

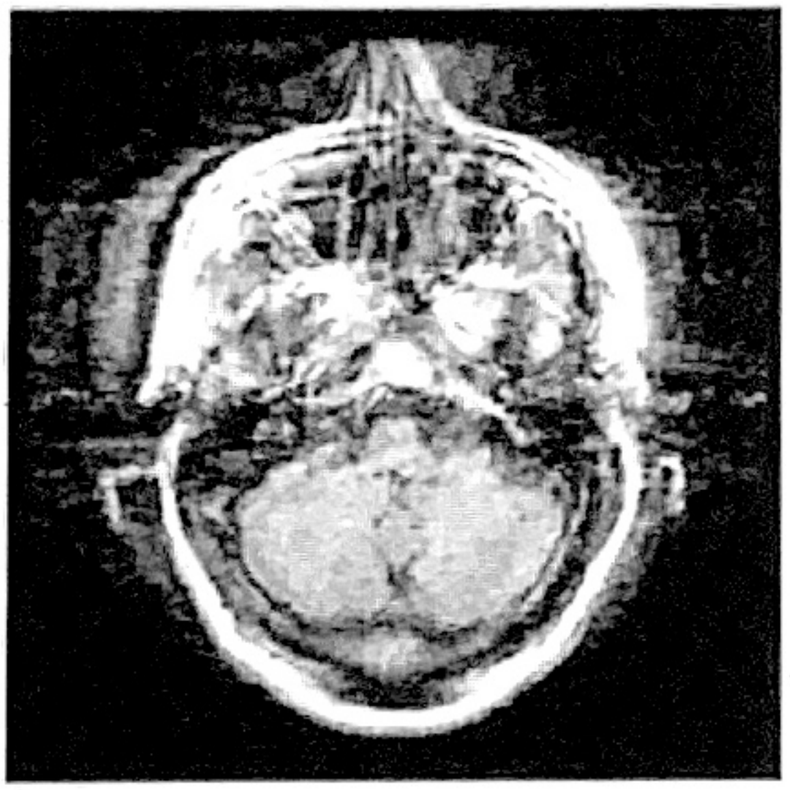


Fig. 4–12: "Motion" artifact

4.4.2 Flow Artifact

Flow can manifest as either altered intravascular signal (flow enhancement or flow-related signal loss), or flow-related artifacts (ghost images or spatial misregistration).

4.4.3 Susceptibility Artifact

Susceptibility artifacts occur at interfaces of tissues with different magnetic susceptibilities. This interface causes B0 distortion, leading to mismapping of the spatial information. This leads to image distortion, peripheral high signal and signal loss.

Although present between different tissues, the effect is much stronger between tissue and metal.

Reduction of these artifacts may be achieved by orientating the long axis of an implant or device parallel to the long axis of the external magnetic field, choosing the appropriate frequency encoding direction, using smaller voxel sizes, using fast imaging sequences, using increased readout bandwidth and avoiding gradient-echo imaging when metal is present.

The image at right shows a susceptibility artifact (left central portion of image), probably due to a surgical clip. The MRA image (bottom) shows almost total loss of signal near the clip.



Fig. 4–13: Susceptibility artifact

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4.4.4 Chemical Shift Artifact

Chemical shift artifact occurs at the fat/water interface in the phase encoding. These artifacts arise due to the difference in resonance of protons resulting from their micromagnetic environment. The protons of fat resonate at a slightly lower frequency than those of water, which causes an offset along the frequency encoding direction.

This artifact typically occurs where anatomy with high water content is adjacent to fat. The image shows a kidney, which has high water density; the dark band on the left and the bright band on the right are chemical shift artifacts.

High field strength magnets are particularly susceptible to this artifact, since the chemical shift is proportional to field strength, while spatial encoding is a function of image bandwidth, which does not scale with field strength.

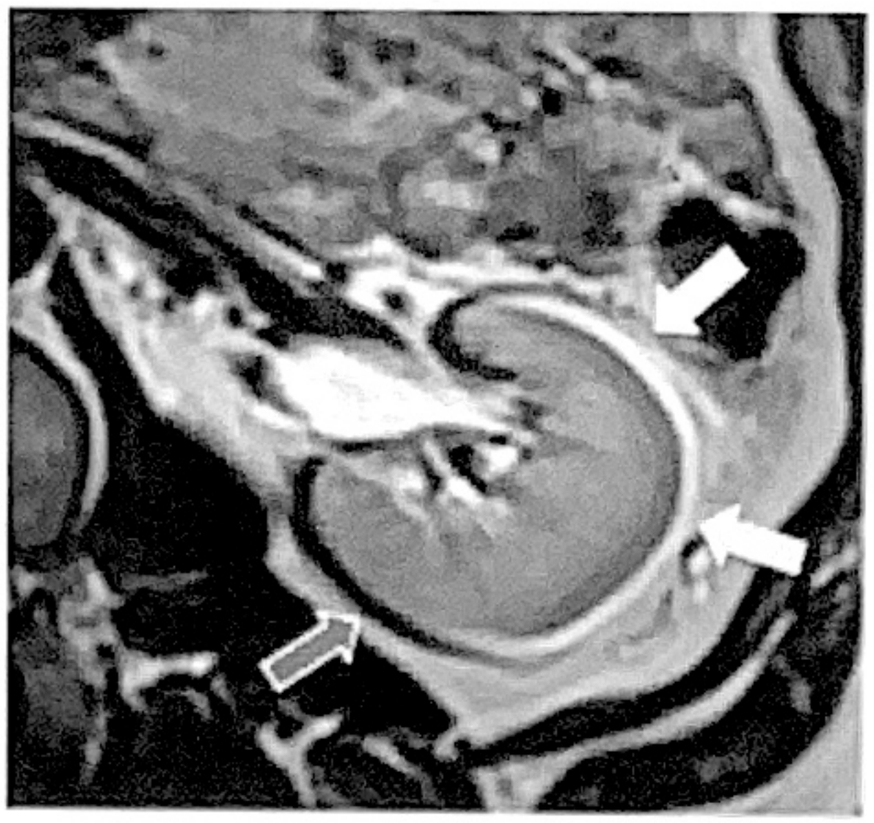


Fig. 4–14: Chem shift artifact

4.4.5 Partial Volume Artifact

Partial volume artifact arises from collecting imaging information of a small object using a voxel size on the same order or smaller than the object. The signal intensity gets averaged with the surrounding tissue in the voxel. To reduce the impact of this artifact, reduce the FOV, increase the matrix size or reduce the slice thickness. Note that any of these actions reduces the SNR.

4.4.6 Phase wrap Artifact

A phase wrap artifact appears as a mismapping of anatomy that lies outside of the image's FOV. This artifact occurs when the FOV used is smaller than the object being imaged. Because a filter is applied in the frequency direction, wrap artifacts occur in the phase-encode direction. To eliminate this artifact, increase the FOV in the phase-encode (PE) direction.

Note that our systems offer a phase-oversampling option to collect a larger FOV in the PE direction but do not display them in order to eliminate this artifact. The downside is increased scan time.

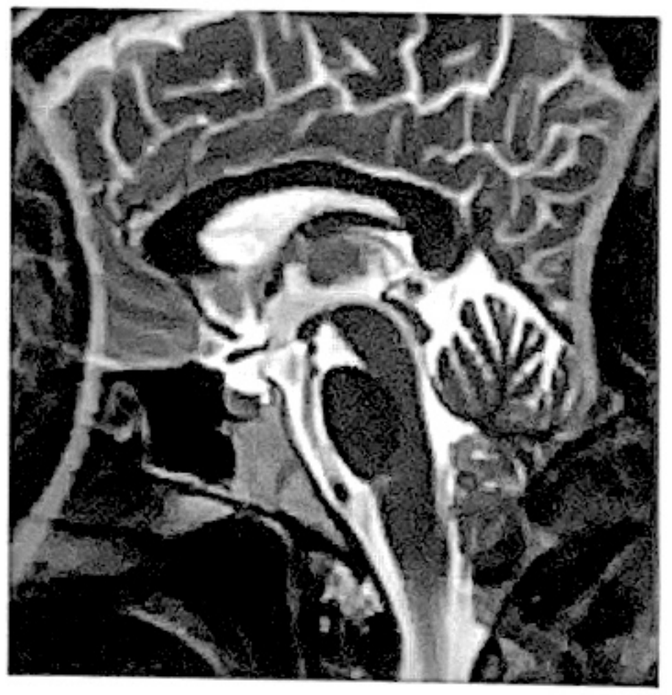


Fig. 4–15: Phase wrap artifact

4.4.7 Gibbs Artifact

The Gibbs artifact, also known as Gibbs ringing, occurs at sharp physical boundaries in the image due to the limited frequency content of the data collected (high-frequency information is required to completely describe a sharp boundary). The artifact appears as alternating light and dark bands (ringing) originating from a sharp boundary. Filtering is applied during reconstruction to reduce this effect.

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4.4.8 Shim Artifact

The fact that the static magnetic field is not perfectly homogeneous over the entire imaging volume leads to shim artifacts. There are two main types of artifacts. First, the magnet designs result is static magnetic fields that have a large high-order term that predominates after the shimming procedure is complete. This causes "bite marks" in images at the boundaries of the imaging volume.

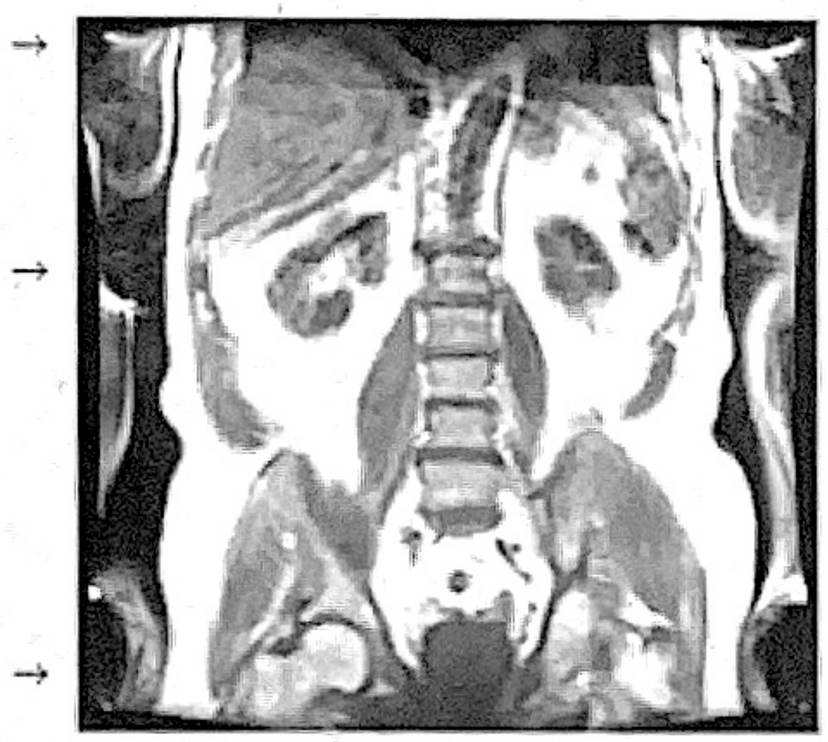


Fig. 4–16: "Bite marks" artifact

In addition, the spectral-saturation method of fat suppression relies on a homogeneous magnetic field. In this method, a frequency-selective pulse is applied to only the fat peak. Since the fat peak frequency differs from the water peak by 3.4ppm, if the static magnet field differs by this much, this method of fat signal reduction is compromised.

This may lead to a reduction in water signal or an image similar to that where fat saturation has not been applied.

Spin-echo sequences are less susceptible to shim artifacts than gradient-echo sequences.

4.4.9 RF Artifact

RF artifacts originate from not applying the desired B1 field at all locations in the imaging volume. The most noticeable affect is image shading. Shading arises from the B1 field in the FOV not being homogeneous and therefore causing the flip angles of the spins to vary as a function of location in space. Variation in flip angle leads to differences in voxel intensities. Due to the relative wavelengths to the human body, this effect is more pronounced at 3T than at 1.5T. Dual drive transmit and RF saturation pads are used to reduce this artifact.

4.4.10 Coherent Noise Artifact

A coherent noise artifact appears as a dot or a noisy line (a "zipper") in the reconstructed image. CN artifacts are caused by fixed frequency noise sources that occurs within the image's bandwidth.

If the noise source is both fixed frequency and phase coherent, the artifact will appear as a bright dot; if not phase coherent, the dot smears into a line along the phase direction. In Fig. 4–17, a vertical zipper implies that frequency is encoded left-to-right.

Even though a CN artifact may be clearly visible in the reconstructed image, it may be difficult or impossible to see in the time domain raw data. In the frequency domain, all the noise energy is mapped into a few pixels, while in the time domain, the same energy is distributed across the entire image.

GEHC's algorithm for detecting CN artifacts is described in DOC0506450, *MRSE Report Coherent Noise Algorithm*.



Fig. 4–17: Coherent noise artifact

4.4.10.1 Rio gradient switching induced body channel coherent noise

During Rio's integration and test, coherent noise was seen on the images using body coil receive due to gradient switching induced transients reaching the body receive preamp's front end. DOC1998819, *Rio Gradient Induced Body Channel Coherent Noise*, covers the investigation and one possible solution.

4.5 Magnet design

MRI systems require a large volume, strong, highly homogeneous, stable B0 field. This field is produced by the magnet. Magnet design and implementation is coordinated with system design, but generally has independent design documentation. An overview is below.

The magnets used in the DV products are

Name	*Series*	*Shimming*	*Used on*
G3	W-series	Supercon	MR750, Rio upgrades
LCC	R-series	Supercon	MR450 (and many previous 1.5T systems)
HDw/DVw	HM-series	Passive	MR450w
3TLC	UM-series	Passive	MR750w, Rio forward production

All of these magnets are actively shielded, cylindrical, superconducting magnets, with B0 along the cylinder axis.

There are two primary methods used to shim the magnet during installation. One option is "supercon" shimming, which uses superconducting shim windings built into the magnet. Located inside magnet's helium vessel, the shim windings are separate from the main superconducting windings (whose function is to create the static magnetic field). An alternative option is "passive" shimming, which uses small pieces of magnetic material placed inside the bore.

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Both of the 60cm bore diameter systems have supercon shimming as a primary method; in addition, there is some limited capability to use passive shim pieces on the MR750 and MR450 systems. Both the 70cm systems (MR450w and MR750w) use only passive shimming; no superconducting shim coils are present.

Cost is the main reason for an all-passive shim design.

4.5.1 Field strength

Higher fields have several advantages. Most importantly, signal strength increases linearly with B0; the number of spins that align with the field is proportional to the field strength, which leads to higher signal. In addition, most T1 and T2 spin relaxation times increase as field strength increases. Longer T1/T2 times lead to less signal decay for MR signals, again leading to stronger signal strength.

Higher fields also have several disadvantages. First, since RF frequency increases with field strength, the energy absorbed by the patient increases, as does B1 transmit non-uniformity. Second, magnet inhomogeneity (which tends to scale with B0) becomes larger compared to imaging bandwidth (which do not necessarily scale with B0). Third, the hazard of magnetic objects being pulled into the magnet increases. Finally, higher fields generally lead to higher static field gradients. The higher field gradients have a greater probability to cause metallic taste, disorientation and/or vomiting.

In general, the advantages outweigh the disadvantages; the commercial success of 3.0T scanners is evidence of the high field advantages.

4.5.2 Homogeneity

The B0 field homogeneity can be expressed as parts per million (PPM) or as Hz (referenced to 1H frequency). At 1.5T, one PPM corresponds 64Hz; at 3.0T, it is 128Hz; therefore, a "1PPM magnet" will have 64Hz inhomogeneity at 1.5T, but have 128Hz at 3T.

Inhomogeneity can be specified by "peak-to-peak" or "RMS" methods. For a peak-to-peak (PP) method, the peak to peak deviation of the magnetic field strength, measured over a defined volume, must be less than the specified value. For the RMS method, the root-mean-square of the field strength is calculated over the defined volume, and compared to the specified value. The PP measurement gives a much higher number than the RMS, so one must be careful to specify the method as well as the expected performance.

Magnet homogeneity often determines what types of applications can be used. Some applications require certain PPM homogeneity. For example, accurate fat saturation (which depends on a highly frequency selective pulse) requires roughly 1PPM homogeneity, since fat is offset from water by about 3.4PPM. Other applications require certain "Hz" homogeneity. Maintaining acceptable EPI image distortions may require 50Hz homogeneity, regardless of B0 strength.

The magnet design dictates the maximum size or volume of homogeneous field; this is referred to as the "FOV" of a magnet. Shimming may improve the homogeneity within this region, but generally cannot extend the volume.

For volumes smaller than the magnet FOV, homogeneity *increases* as the measurement volume *decreases*, for a given magnet design. The homogeneity on a 12cm FOV may be 10X better than the inhomogeneity at 36cm, which in turn may be 10X better than at 48cm. Therefore, fat saturation may work well on small FOV scan, but fail on larger FOV scans (which is often were it is needed most.)

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The table below summarizes the different systems' homogeneity, in rms PPM over the specified volume.

DSV, cm [1]	MR450	MR750	MR450w	MR750w
50	2.66		3.3	2.22
48	1.66		2.3 [2]	1.71 [2]
45	0.81		1.2	0.68
40	0.27	0.25	0.50	0.23
35	0.116			0.10
30	0.062	0.10	0.11	0.06
20	0.020	0.05	0.035	0.021
10	0.004	0.03	0.007	0.005

[1] DSV is a spherical volume of the given diameter.
[2] MR450w and MR750w systems use 50x50x45 cm ellipsoidal volume, with the short axis along z.

4.5.3 Stability

The stability of the homogeneous volume over time is also critical to image quality. The magnet design has a direct impact of all three types of field drift (refer to section **Error! Reference source not found.**). Design elements such as quench circuit, internal mechanical connections, coldhead location and total weight factor into the field stability performance.

(To be expanded) HSS

4.5.4 Safety

- Bloom field
- 5G line
- Maximum gradient field

4.6 The gradient subsystem

General information on MRI gradient fields is in section 3.5 *The gradient field*; section 3.7 *The gradient subsystem* goes further into the subsystem design. Section 3.7.2 *Current and voltage waveforms* gives basic theory on gradient waveforms.

The GEHC premium systems use the XGD or XG2 gradient drivers and the XRMB and XRMw gradient coils. A block diagram of the gradient subsystem is in Fig. 4–18.

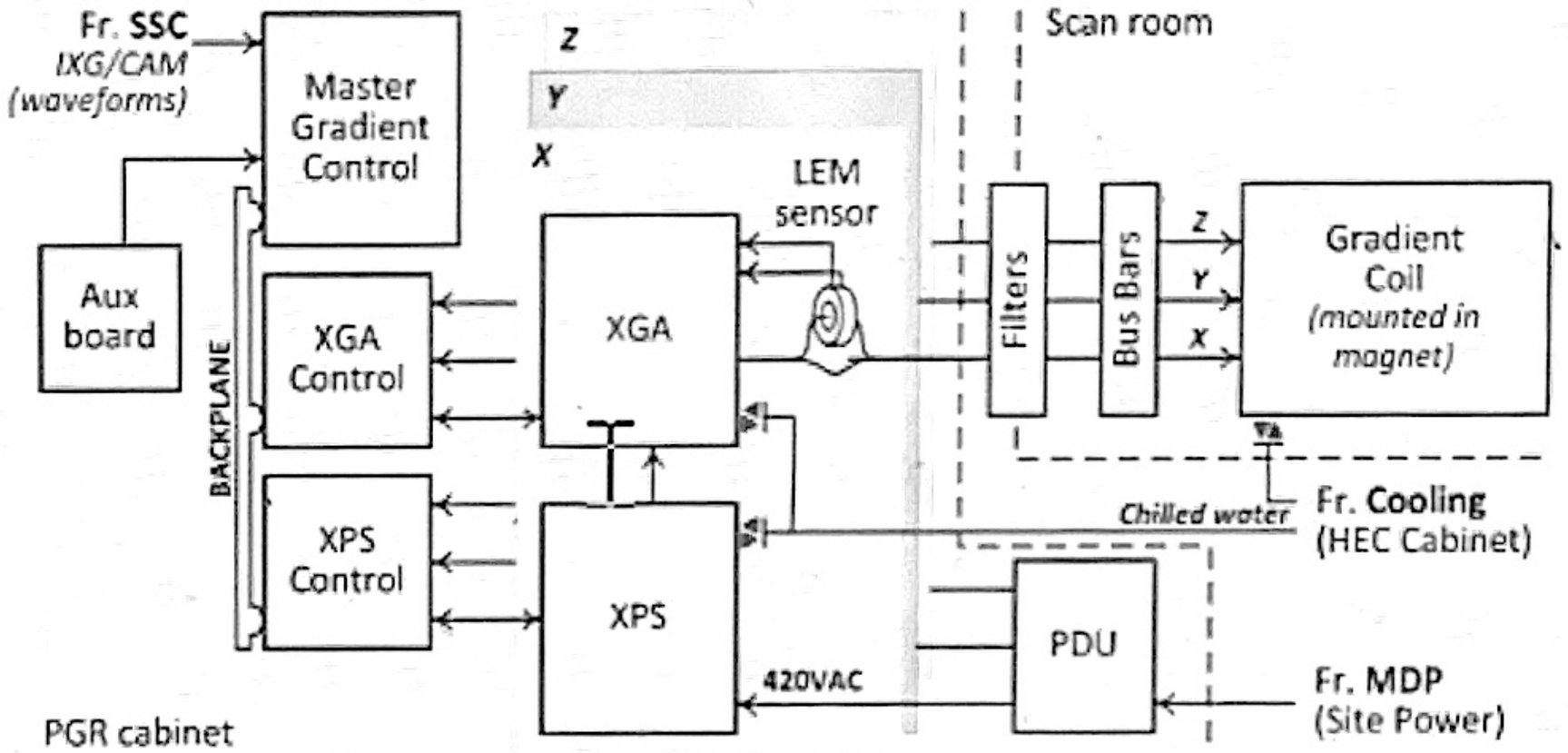


Fig. 4–18: The XGD/XRM gradient subsystem

The gradient subsystem provides gradient fields to the system. The main components for this gradient subsystem are

1) Gradient drivers (controls, XGA amplifiers, XPS power supplies)
2) Gradient coil (XRMB or XRMw)
3) Gradient cables, filters, busbars
4) PDU (AC power distribution unit)

The gradient driver consists of the controllers, amplifiers, and power supplies.

The MGC (Master Gradient Controller) provides a system interface, accepting 1) waveform data from the SSC subsystem, and 2) control and command information. The XGA control and XPS control provide support for the XGA and XPS, respectively.

The three XPS (X-treme Power Supply) are high efficiency units that convert 420VAC 3ϕ power into DC power for use by the XGA.

The three XGAs (X-treme Gradient Amplifiers) are IGBT based switching amplifiers, converting the DC power from the XPS into gradient current. This output current is monitored by the LEMs, highly accurate and low noise current transducers, which provide feedback to the XGAs.

The gradient cables route the gradient currents from the PGR cabinet output to the gradient coil. To provide mechanical flexibility combined with high current capacity, four conductor cables are used; two conductors each for the plus and minus signals.

The gradient current is carried thru the gradient filters, mounted on the penetration panel. The filters block RF noise from passing into the scan room; they do not significantly filter the switching ripple on the gradient lines.

The magnet bus-bars route the gradient current as it passes near the magnet. The magnetic field in this region generates large forces on the conductors, which the bars are designed to handle. (The "busbar" system safety limit protects a different set of bus-bars, which are located inside the XGAs.)

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